

ANALYTICAL REPORT

Eurofins TestAmerica, Michigan
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Laboratory Job ID: 190-24417-1

Client Project/Site: City of Cadillac Semiannual PFAS

For:

City of Cadillac Utilities
1121 Plett Road
Cadillac, Michigan 49601

Attn: Cindy Tomaszewski

Sue Schafer

Authorized for release by:
10/19/2020 4:48:47 PM

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Results relate only to the items tested and the sample(s) as received by the laboratory.



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Sample Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-24417-1

Lab Sample ID	Client Sample ID	Matrix	Collected	Received	Asset ID
190-24417-1	Effluent	Water	10/07/20 11:28	10/08/20 12:10	
190-24417-2	Effluent Duplic.	Water	10/07/20 11:30	10/08/20 12:10	
190-24417-3	Field Blank	Water	10/07/20 11:25	10/08/20 12:10	

Case Narrative

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-24417-1

Job ID: 190-24417-1

Laboratory: Eurofins TestAmerica, Michigan

Narrative

Job Narrative 190-24417-1

Comments

No additional comments.

Receipt

The samples were received on 10/8/2020 12:10 PM; the samples arrived in good condition, and where required, properly preserved and on ice. The temperature of the cooler at receipt was 3.5° C.

LCMS

Method 537 (modified): Compound 13C2 PFOA eluted outside the retention time window on the C-18 column for the following sample: Effluent (190-24417-1). This retention time shift was taken into account when reviewing the sample for target compounds.

Method 537 (modified): The laboratory control sample (LCS) for preparation batch 200-159844 and analytical batch 200-159860 recovered outside control limits for the following analytes: Perfluorotetradecanoic acid (PFTeA). This analyte was biased high in the LCS and was not detected in the associated samples; therefore, the data have been reported.

Method 537 (modified): 13C2 PFTeDA Isotope Dilution Analyte (IDA) recovery associated with the following samples is below the method recommended limit: (LCS 200-159844/2-A) and (MB 200-159844/1-A). Generally, data quality is not considered affected if the IDA signal-to-noise ratio is greater than 10:1, which is achieved for all IDA in the sample(s).

Method 537 (modified): The continuing calibration verification (CCV) associated with batch 200-159858 recovered above the upper control limit for 13C3 HFPO-DA. The samples associated with this CCV were non-detects for the affected analytes; therefore, the data have been reported.

No additional analytical or quality issues were noted, other than those described above or in the Definitions/Glossary page.

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-24417-1

Client Sample ID: Effluent

Lab Sample ID: 190-24417-1

Date Collected: 10/07/20 11:28

Matrix: Water

Date Received: 10/08/20 12:10

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
11-Chloroeicosafluoro-3-oxaundecan e-1-sulfonic acid	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:01	1
9-Chlorohexadecafluoro-3-oxanonan e-1-sulfonic acid	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:01	1
DONA	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:01	1
4:2 FTS	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:01	1
6:2 FTS	<4.2		4.2	ng/L		10/12/20 13:33	10/12/20 22:01	1
8:2 FTS	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:01	1
HFPO-DA	<3.4		3.4	ng/L		10/12/20 13:33	10/12/20 22:01	1
N-ethylperfluorooctanesulfonamidoac etic acid (NEtFOSAA)	<4.2		4.2	ng/L		10/12/20 13:33	10/12/20 22:01	1
N-methylperfluorooctanesulfonamidoa cetic acid (NMeFOSAA)	<4.2		4.2	ng/L		10/12/20 13:33	10/12/20 22:01	1
Perfluorobutanesulfonic acid (PFBS)	25		1.7	ng/L		10/12/20 13:33	10/12/20 22:01	1
Perfluorobutanoic acid (PFBA)	<4.2		4.2	ng/L		10/12/20 13:33	10/12/20 22:01	1
Perfluorodecanesulfonic acid (PFDS)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:01	1
Perfluorodecanoic acid (PFDA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:01	1
Perfluorododecanoic acid (PFDoA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:01	1
Perfluoroheptanesulfonic Acid (PFHpS)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:01	1
Perfluoroheptanoic acid (PFHpA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:01	1
Perfluorohexanesulfonic acid (PFHxS)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:01	1
Perfluorohexanoic acid (PFHxA)	9.6		1.7	ng/L		10/12/20 13:33	10/12/20 22:01	1
Perfluorononanesulfonic acid (PFNS)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:01	1
Perfluorononanoic acid (PFNA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:01	1
Perfluorooctanesulfonamide (FOSA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:01	1
Perfluorooctane sulfonate (PFOS)	1.8		1.7	ng/L		10/12/20 13:33	10/12/20 22:01	1
Perfluorooctanoic acid (PFOA)	2.5		1.7	ng/L		10/12/20 13:33	10/12/20 22:01	1
Perfluoropentanesulfonic acid (PFPeS)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:01	1
Perfluoropentanoic acid (PFPeA)	3.4		1.7	ng/L		10/12/20 13:33	10/12/20 22:01	1
Perfluorotetradecanoic acid (PFTeA)	<1.7 *		1.7	ng/L		10/12/20 13:33	10/12/20 22:01	1
Perfluorotridecanoic acid (PFTriA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:01	1
Perfluoroundecanoic acid (PFUnA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:01	1

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	66		25 - 150	10/12/20 13:33	10/12/20 22:01	1
13C3 HFPO-DA	101		25 - 150	10/12/20 13:33	10/12/20 22:01	1
13C4 PFBA	114		25 - 150	10/12/20 13:33	10/12/20 22:01	1
13C3 PFBS	82		50 - 150	10/12/20 13:33	10/12/20 22:01	1
13C2 PFDA	76		50 - 150	10/12/20 13:33	10/12/20 22:01	1
13C2 PFDoA	57		50 - 150	10/12/20 13:33	10/12/20 22:01	1
13C4 PFHpA	88		50 - 150	10/12/20 13:33	10/12/20 22:01	1
13C2 PFHxA	88		50 - 150	10/12/20 13:33	10/12/20 22:01	1
13C5 PFNA	80		50 - 150	10/12/20 13:33	10/12/20 22:01	1
13C4 PFOA	84		50 - 150	10/12/20 13:33	10/12/20 22:01	1
13C4 PFOS	81		50 - 150	10/12/20 13:33	10/12/20 22:01	1
13C5 PFPeA	95		25 - 150	10/12/20 13:33	10/12/20 22:01	1
13C2 PFTeDA	59		50 - 150	10/12/20 13:33	10/12/20 22:01	1
13C2 PFUnA	74		50 - 150	10/12/20 13:33	10/12/20 22:01	1
d5-NEtFOSAA	63		50 - 150	10/12/20 13:33	10/12/20 22:01	1

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Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-24417-1

Client Sample ID: Effluent

Date Collected: 10/07/20 11:28

Date Received: 10/08/20 12:10

Lab Sample ID: 190-24417-1

Matrix: Water

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

<i>Isotope Dilution</i>	<i>%Recovery</i>	<i>Qualifier</i>	<i>Limits</i>	<i>Prepared</i>	<i>Analyzed</i>	<i>Dil Fac</i>
d3-NMeFOSAA	62		50 - 150	10/12/20 13:33	10/12/20 22:01	1
M2-4:2 FTS	123		25 - 150	10/12/20 13:33	10/12/20 22:01	1
M2-6:2 FTS	105		25 - 150	10/12/20 13:33	10/12/20 22:01	1
M2-8:2 FTS	102		25 - 150	10/12/20 13:33	10/12/20 22:01	1
18O2 PFHxS	87		50 - 150	10/12/20 13:33	10/12/20 22:01	1

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-24417-1

Client Sample ID: Effluent Duplic.

Lab Sample ID: 190-24417-2

Date Collected: 10/07/20 11:30

Matrix: Water

Date Received: 10/08/20 12:10

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
11-Chloroeicosafluoro-3-oxaundecan e-1-sulfonic acid	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:09	1
9-Chlorohexadecafluoro-3-oxanonan e-1-sulfonic acid	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:09	1
DONA	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:09	1
4:2 FTS	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:09	1
6:2 FTS	<4.2		4.2	ng/L		10/12/20 13:33	10/12/20 22:09	1
8:2 FTS	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:09	1
HFPO-DA	<3.3		3.3	ng/L		10/12/20 13:33	10/12/20 22:09	1
N-ethylperfluorooctanesulfonamidoac etic acid (NEtFOSAA)	<4.2		4.2	ng/L		10/12/20 13:33	10/12/20 22:09	1
N-methylperfluorooctanesulfonamidoa cetic acid (NMeFOSAA)	<4.2		4.2	ng/L		10/12/20 13:33	10/12/20 22:09	1
Perfluorobutanesulfonic acid (PFBS)	25		1.7	ng/L		10/12/20 13:33	10/12/20 22:09	1
Perfluorobutanoic acid (PFBA)	<4.2		4.2	ng/L		10/12/20 13:33	10/12/20 22:09	1
Perfluorodecanesulfonic acid (PFDS)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:09	1
Perfluorodecanoic acid (PFDA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:09	1
Perfluorododecanoic acid (PFDoA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:09	1
Perfluoroheptanesulfonic Acid (PFHpS)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:09	1
Perfluoroheptanoic acid (PFHpA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:09	1
Perfluorohexanesulfonic acid (PFHxS)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:09	1
Perfluorohexanoic acid (PFHxA)	9.5		1.7	ng/L		10/12/20 13:33	10/12/20 22:09	1
Perfluorononanesulfonic acid (PFNS)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:09	1
Perfluorononanoic acid (PFNA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:09	1
Perfluorooctanesulfonamide (FOSA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:09	1
Perfluorooctane sulfonate (PFOS)	1.8		1.7	ng/L		10/12/20 13:33	10/12/20 22:09	1
Perfluorooctanoic acid (PFOA)	2.4		1.7	ng/L		10/12/20 13:33	10/12/20 22:09	1
Perfluoropentanesulfonic acid (PFPeS)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:09	1
Perfluoropentanoic acid (PFPeA)	3.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:09	1
Perfluorotetradecanoic acid (PFTeA)	<1.7 *		1.7	ng/L		10/12/20 13:33	10/12/20 22:09	1
Perfluorotridecanoic acid (PFTriA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:09	1
Perfluoroundecanoic acid (PFUnA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:09	1

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	61		25 - 150	10/12/20 13:33	10/12/20 22:09	1
13C3 HFPO-DA	119		25 - 150	10/12/20 13:33	10/12/20 22:09	1
13C4 PFBA	109		25 - 150	10/12/20 13:33	10/12/20 22:09	1
13C3 PFBS	80		50 - 150	10/12/20 13:33	10/12/20 22:09	1
13C2 PFDA	72		50 - 150	10/12/20 13:33	10/12/20 22:09	1
13C2 PFDoA	52		50 - 150	10/12/20 13:33	10/12/20 22:09	1
13C4 PFHpA	90		50 - 150	10/12/20 13:33	10/12/20 22:09	1
13C2 PFHxA	92		50 - 150	10/12/20 13:33	10/12/20 22:09	1
13C5 PFNA	73		50 - 150	10/12/20 13:33	10/12/20 22:09	1
13C4 PFOA	83		50 - 150	10/12/20 13:33	10/12/20 22:09	1
13C4 PFOS	78		50 - 150	10/12/20 13:33	10/12/20 22:09	1
13C5 PFPeA	90		25 - 150	10/12/20 13:33	10/12/20 22:09	1
13C2 PFTeDA	65		50 - 150	10/12/20 13:33	10/12/20 22:09	1
13C2 PFUnA	75		50 - 150	10/12/20 13:33	10/12/20 22:09	1
d5-NEtFOSAA	58		50 - 150	10/12/20 13:33	10/12/20 22:09	1

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Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-24417-1

Client Sample ID: Effluent Duplic.

Lab Sample ID: 190-24417-2

Date Collected: 10/07/20 11:30

Matrix: Water

Date Received: 10/08/20 12:10

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

<i>Isotope Dilution</i>	<i>%Recovery</i>	<i>Qualifier</i>	<i>Limits</i>	<i>Prepared</i>	<i>Analyzed</i>	<i>Dil Fac</i>
d3-NMeFOSAA	65		50 - 150	10/12/20 13:33	10/12/20 22:09	1
M2-4:2 FTS	121		25 - 150	10/12/20 13:33	10/12/20 22:09	1
M2-6:2 FTS	116		25 - 150	10/12/20 13:33	10/12/20 22:09	1
M2-8:2 FTS	100		25 - 150	10/12/20 13:33	10/12/20 22:09	1
18O2 PFHxS	87		50 - 150	10/12/20 13:33	10/12/20 22:09	1

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-24417-1

Client Sample ID: Field Blank

Lab Sample ID: 190-24417-3

Date Collected: 10/07/20 11:25

Matrix: Water

Date Received: 10/08/20 12:10

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
11-Chloroeicosafluoro-3-oxaundecan e-1-sulfonic acid	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:17	1
9-Chlorohexadecafluoro-3-oxanonan e-1-sulfonic acid	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:17	1
DONA	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:17	1
4:2 FTS	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:17	1
6:2 FTS	<4.3		4.3	ng/L		10/12/20 13:33	10/12/20 22:17	1
8:2 FTS	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:17	1
HFPO-DA	<3.4		3.4	ng/L		10/12/20 13:33	10/12/20 22:17	1
N-ethylperfluorooctanesulfonamidoac etic acid (NEtFOSAA)	<4.3		4.3	ng/L		10/12/20 13:33	10/12/20 22:17	1
N-methylperfluorooctanesulfonamidoa cetic acid (NMeFOSAA)	<4.3		4.3	ng/L		10/12/20 13:33	10/12/20 22:17	1
Perfluorobutanesulfonic acid (PFBS)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:17	1
Perfluorobutanoic acid (PFBA)	<4.3		4.3	ng/L		10/12/20 13:33	10/12/20 22:17	1
Perfluorodecanesulfonic acid (PFDS)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:17	1
Perfluorodecanoic acid (PFDA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:17	1
Perfluorododecanoic acid (PFDoA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:17	1
Perfluoroheptanesulfonic Acid (PFHpS)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:17	1
Perfluoroheptanoic acid (PFHpA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:17	1
Perfluorohexanesulfonic acid (PFHxS)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:17	1
Perfluorohexanoic acid (PFHxA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:17	1
Perfluorononanesulfonic acid (PFNS)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:17	1
Perfluorononanoic acid (PFNA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:17	1
Perfluorooctanesulfonamide (FOSA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:17	1
Perfluorooctane sulfonate (PFOS)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:17	1
Perfluorooctanoic acid (PFOA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:17	1
Perfluoropentanesulfonic acid (PFPeS)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:17	1
Perfluoropentanoic acid (PFPeA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:17	1
Perfluorotetradecanoic acid (PFTeA)	<1.7 *		1.7	ng/L		10/12/20 13:33	10/12/20 22:17	1
Perfluorotridecanoic acid (PFTriA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:17	1
Perfluoroundecanoic acid (PFUnA)	<1.7		1.7	ng/L		10/12/20 13:33	10/12/20 22:17	1

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	66		25 - 150	10/12/20 13:33	10/12/20 22:17	1
13C3 HFPO-DA	106		25 - 150	10/12/20 13:33	10/12/20 22:17	1
13C4 PFBA	108		25 - 150	10/12/20 13:33	10/12/20 22:17	1
13C3 PFBS	92		50 - 150	10/12/20 13:33	10/12/20 22:17	1
13C2 PFDA	80		50 - 150	10/12/20 13:33	10/12/20 22:17	1
13C2 PFDoA	62		50 - 150	10/12/20 13:33	10/12/20 22:17	1
13C4 PFHpA	96		50 - 150	10/12/20 13:33	10/12/20 22:17	1
13C2 PFHxA	99		50 - 150	10/12/20 13:33	10/12/20 22:17	1
13C5 PFNA	90		50 - 150	10/12/20 13:33	10/12/20 22:17	1
13C4 PFOA	88		50 - 150	10/12/20 13:33	10/12/20 22:17	1
13C4 PFOS	93		50 - 150	10/12/20 13:33	10/12/20 22:17	1
13C5 PFPeA	99		25 - 150	10/12/20 13:33	10/12/20 22:17	1
13C2 PFTeDA	66		50 - 150	10/12/20 13:33	10/12/20 22:17	1
13C2 PFUnA	81		50 - 150	10/12/20 13:33	10/12/20 22:17	1
d5-NEtFOSAA	77		50 - 150	10/12/20 13:33	10/12/20 22:17	1
d3-NMeFOSAA	75		50 - 150	10/12/20 13:33	10/12/20 22:17	1

Eurofins TestAmerica, Michigan

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-24417-1

Client Sample ID: Field Blank

Lab Sample ID: 190-24417-3

Date Collected: 10/07/20 11:25

Matrix: Water

Date Received: 10/08/20 12:10

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

<i>Isotope Dilution</i>	<i>%Recovery</i>	<i>Qualifier</i>	<i>Limits</i>	<i>Prepared</i>	<i>Analyzed</i>	<i>Dil Fac</i>
M2-4:2 FTS	110		25 - 150	10/12/20 13:33	10/12/20 22:17	1
M2-6:2 FTS	104		25 - 150	10/12/20 13:33	10/12/20 22:17	1
M2-8:2 FTS	102		25 - 150	10/12/20 13:33	10/12/20 22:17	1
18O2 PFHxS	103		50 - 150	10/12/20 13:33	10/12/20 22:17	1

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-24417-1

Method: 537 (modified) - Fluorinated Alkyl Substances

Lab Sample ID: MB 200-159844/1-A

Matrix: Water

Analysis Batch: 159860

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 159844

Analyte	MB Result	MB Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
11-Chloroeicosafluoro-3-oxaundecan e-1-sulfonic acid	<2.0		2.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
9-Chlorohexadecafluoro-3-oxanonan e-1-sulfonic acid	<2.0		2.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
DONA	<2.0		2.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
4:2 FTS	<2.0		2.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
6:2 FTS	<5.0		5.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
8:2 FTS	<2.0		2.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
HFPO-DA	<4.0		4.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
N-ethylperfluorooctanesulfonamidoac etic acid (NEtFOSAA)	<5.0		5.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
N-methylperfluorooctanesulfonamidoa cetic acid (NMeFOSAA)	<5.0		5.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
Perfluorobutanesulfonic acid (PFBS)	<2.0		2.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
Perfluorobutanoic acid (PFBA)	<5.0		5.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
Perfluorodecanesulfonic acid (PFDS)	<2.0		2.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
Perfluorodecanoic acid (PFDA)	<2.0		2.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
Perfluorododecanoic acid (PFDoA)	<2.0		2.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
Perfluoroheptanesulfonic Acid (PFHpS)	<2.0		2.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
Perfluoroheptanoic acid (PFHpA)	<2.0		2.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
Perfluorohexanesulfonic acid (PFHxS)	<2.0		2.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
Perfluorohexanoic acid (PFHxA)	<2.0		2.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
Perfluorononanesulfonic acid (PFNS)	<2.0		2.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
Perfluorononanoic acid (PFNA)	<2.0		2.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
Perfluorooctanesulfonamide (FOSA)	<2.0		2.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
Perfluorooctane sulfonate (PFOS)	<2.0		2.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
Perfluorooctanoic acid (PFOA)	<2.0		2.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
Perfluoropentanesulfonic acid (PFPeS)	<2.0		2.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
Perfluoropentanoic acid (PFPeA)	<2.0		2.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
Perfluorotetradecanoic acid (PFTeA)	<2.0		2.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
Perfluorotridecanoic acid (PFTrIA)	<2.0		2.0	ng/L		10/12/20 13:33	10/12/20 20:46	1
Perfluoroundecanoic acid (PFUnA)	<2.0		2.0	ng/L		10/12/20 13:33	10/12/20 20:46	1

Isotope Dilution	MB %Recovery	MB Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	78		25 - 150	10/12/20 13:33	10/12/20 20:46	1
13C3 HFPO-DA	115		25 - 150	10/12/20 13:33	10/12/20 20:46	1
13C4 PFBA	93		25 - 150	10/12/20 13:33	10/12/20 20:46	1
13C3 PFBS	90		50 - 150	10/12/20 13:33	10/12/20 20:46	1
13C2 PFDA	84		50 - 150	10/12/20 13:33	10/12/20 20:46	1
13C2 PFDoA	59		50 - 150	10/12/20 13:33	10/12/20 20:46	1
13C4 PFHpA	89		50 - 150	10/12/20 13:33	10/12/20 20:46	1
13C2 PFHxA	97		50 - 150	10/12/20 13:33	10/12/20 20:46	1
13C5 PFNA	81		50 - 150	10/12/20 13:33	10/12/20 20:46	1
13C4 PFOA	86		50 - 150	10/12/20 13:33	10/12/20 20:46	1
13C4 PFOS	91		50 - 150	10/12/20 13:33	10/12/20 20:46	1
13C5 PFPeA	94		25 - 150	10/12/20 13:33	10/12/20 20:46	1
13C2 PFTeDA	49	*5	50 - 150	10/12/20 13:33	10/12/20 20:46	1
13C2 PFUnA	80		50 - 150	10/12/20 13:33	10/12/20 20:46	1

Eurofins TestAmerica, Michigan

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-24417-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: MB 200-159844/1-A

Matrix: Water

Analysis Batch: 159860

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 159844

Isotope Dilution	MB %Recovery	MB Qualifier	Limits	Prepared	Analyzed	Dil Fac
d5-NEtFOSAA	77		50 - 150	10/12/20 13:33	10/12/20 20:46	1
d3-NMeFOSAA	72		50 - 150	10/12/20 13:33	10/12/20 20:46	1
M2-4:2 FTS	102		25 - 150	10/12/20 13:33	10/12/20 20:46	1
M2-6:2 FTS	98		25 - 150	10/12/20 13:33	10/12/20 20:46	1
M2-8:2 FTS	96		25 - 150	10/12/20 13:33	10/12/20 20:46	1
18O2 PFHxS	92		50 - 150	10/12/20 13:33	10/12/20 20:46	1

Lab Sample ID: LCS 200-159844/2-A

Matrix: Water

Analysis Batch: 159860

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 159844

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
11-Chloroeicosafluoro-3-oxaundecane-1-sulfonic acid	37.7	26.2		ng/L		70	50 - 150
9-Chlorohexadecafluoro-3-oxanonane-1-sulfonic acid	37.3	37.0		ng/L		99	50 - 150
DONA	37.7	35.9		ng/L		95	50 - 150
4:2 FTS	37.4	36.5		ng/L		98	50 - 150
6:2 FTS	37.9	37.5		ng/L		99	50 - 150
8:2 FTS	38.3	39.0		ng/L		102	50 - 150
HFPO-DA	40.0	33.3		ng/L		83	50 - 150
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	40.0	34.4		ng/L		86	70 - 130
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	40.0	40.7		ng/L		102	70 - 130
Perfluorobutanesulfonic acid (PFBS)	35.4	40.9		ng/L		116	70 - 130
Perfluorobutanoic acid (PFBA)	40.0	39.5		ng/L		99	50 - 150
Perfluorodecanesulfonic acid (PFDS)	38.6	31.9		ng/L		83	50 - 150
Perfluorodecanoic acid (PFDA)	40.0	40.6		ng/L		101	70 - 130
Perfluorododecanoic acid (PFDoA)	40.0	44.1		ng/L		110	70 - 130
Perfluoroheptanesulfonic Acid (PFHpS)	38.1	39.3		ng/L		103	50 - 150
Perfluoroheptanoic acid (PFHpA)	40.0	42.0		ng/L		105	70 - 130
Perfluorohexanesulfonic acid (PFHxS)	36.4	35.1		ng/L		96	70 - 130
Perfluorohexanoic acid (PFHxA)	40.0	40.6		ng/L		102	70 - 130
Perfluorononanesulfonic acid (PFNS)	38.4	37.4		ng/L		97	50 - 150
Perfluorononanoic acid (PFNA)	40.0	40.3		ng/L		101	70 - 130
Perfluorooctanesulfonamide (FOSA)	40.0	43.2		ng/L		108	50 - 150
Perfluorooctane sulfonate (PFOS)	37.1	36.5		ng/L		98	70 - 130
Perfluorooctanoic acid (PFOA)	40.0	42.2		ng/L		105	70 - 130
Perfluoropentanesulfonic acid (PFPeS)	37.5	39.9		ng/L		106	50 - 150
Perfluoropentanoic acid (PFPeA)	40.0	37.2		ng/L		93	50 - 150
Perfluorotetradecanoic acid (PFTeA)	40.0	53.3 *		ng/L		133	70 - 130

Eurofins TestAmerica, Michigan

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-24417-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: LCS 200-159844/2-A

Matrix: Water

Analysis Batch: 159860

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 159844

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
Perfluorotridecanoic acid (PFTriA)	40.0	37.7		ng/L		94	70 - 130
Perfluoroundecanoic acid (PFUnA)	40.0	39.4		ng/L		98	70 - 130

Isotope Dilution	LCS %Recovery	LCS Qualifier	Limits
13C8 FOSA	61		25 - 150
13C3 HFPO-DA	114		25 - 150
13C4 PFBA	101		25 - 150
13C3 PFBS	90		50 - 150
13C2 PFDA	81		50 - 150
13C2 PFDoA	50		50 - 150
13C4 PFHpA	96		50 - 150
13C2 PFHxA	99		50 - 150
13C5 PFNA	84		50 - 150
13C4 PFOA	86		50 - 150
13C4 PFOS	94		50 - 150
13C5 PFPeA	101		25 - 150
13C2 PFTeDA	44	*5	50 - 150
13C2 PFUnA	82		50 - 150
d5-NEtFOSAA	82		50 - 150
d3-NMeFOSAA	78		50 - 150
M2-4:2 FTS	103		25 - 150
M2-6:2 FTS	96		25 - 150
M2-8:2 FTS	96		25 - 150
18O2 PFHxS	99		50 - 150

Isotope Dilution Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-24417-1

Method: 537 (modified) - Fluorinated Alkyl Substances

Matrix: Water

Prep Type: Total/NA

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	PFOSA (25-150)	HFPODA (25-150)	PFBA (25-150)	C3PFBS (50-150)	PFDA (50-150)	PFDoA (50-150)	C4PFHA (50-150)	PFHxA (50-150)
190-24417-1	Effluent	66	101	114	82	76	57	88	88
190-24417-2	Effluent Duplic.	61	119	109	80	72	52	90	92
190-24417-3	Field Blank	66	106	108	92	80	62	96	99
LCS 200-159844/2-A	Lab Control Sample	61	114	101	90	81	50	96	99
MB 200-159844/1-A	Method Blank	78	115	93	90	84	59	89	97

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	PFNA (50-150)	PFOA (50-150)	PFOS (50-150)	PFPeA (25-150)	PFTDA (50-150)	PFUnA (50-150)	d5NEFOS (50-150)	d3NMFOS (50-150)
190-24417-1	Effluent	80	84	81	95	59	74	63	62
190-24417-2	Effluent Duplic.	73	83	78	90	65	75	58	65
190-24417-3	Field Blank	90	88	93	99	66	81	77	75
LCS 200-159844/2-A	Lab Control Sample	84	86	94	101	44 *5	82	82	78
MB 200-159844/1-A	Method Blank	81	86	91	94	49 *5	80	77	72

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	M242FTS (25-150)	M262FTS (25-150)	M282FTS (25-150)	PFHxS (50-150)
190-24417-1	Effluent	123	105	102	87
190-24417-2	Effluent Duplic.	121	116	100	87
190-24417-3	Field Blank	110	104	102	103
LCS 200-159844/2-A	Lab Control Sample	103	96	96	99
MB 200-159844/1-A	Method Blank	102	98	96	92

Surrogate Legend

PFOSA = 13C8 FOSA
HFPODA = 13C3 HFPO-DA
PFBA = 13C4 PFBA
C3PFBS = 13C3 PFBS
PFDA = 13C2 PFDA
PFDoA = 13C2 PFDoA
C4PFHA = 13C4 PFHpA
PFHxA = 13C2 PFHxA
PFNA = 13C5 PFNA
PFOA = 13C4 PFOA
PFOS = 13C4 PFOS
PFPeA = 13C5 PFPeA
PFTDA = 13C2 PFTeDA
PFUnA = 13C2 PFUnA
d5NEFOS = d5-NEtFOSAA
d3NMFOS = d3-NMeFOSAA
M242FTS = M2-4:2 FTS
M262FTS = M2-6:2 FTS
M282FTS = M2-8:2 FTS
PFHxS = 18O2 PFHxS

Definitions/Glossary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-24417-1

Qualifiers

LCMS

Qualifier	Qualifier Description
*	LCS or LCSD is outside acceptance limits.
*5	Isotope dilution analyte is outside acceptance limits.

Glossary

Abbreviation	These commonly used abbreviations may or may not be present in this report.
α	Listed under the "D" column to designate that the result is reported on a dry weight basis
%R	Percent Recovery
CFL	Contains Free Liquid
CFU	Colony Forming Unit
CNF	Contains No Free Liquid
DER	Duplicate Error Ratio (normalized absolute difference)
Dil Fac	Dilution Factor
DL	Detection Limit (DoD/DOE)
DL, RA, RE, IN	Indicates a Dilution, Re-analysis, Re-extraction, or additional Initial metals/anion analysis of the sample
DLC	Decision Level Concentration (Radiochemistry)
EDL	Estimated Detection Limit (Dioxin)
LOD	Limit of Detection (DoD/DOE)
LOQ	Limit of Quantitation (DoD/DOE)
MCL	EPA recommended "Maximum Contaminant Level"
MDA	Minimum Detectable Activity (Radiochemistry)
MDC	Minimum Detectable Concentration (Radiochemistry)
MDL	Method Detection Limit
ML	Minimum Level (Dioxin)
MPN	Most Probable Number
MQL	Method Quantitation Limit
NC	Not Calculated
ND	Not Detected at the reporting limit (or MDL or EDL if shown)
NEG	Negative / Absent
POS	Positive / Present
PQL	Practical Quantitation Limit
PRES	Presumptive
QC	Quality Control
RER	Relative Error Ratio (Radiochemistry)
RL	Reporting Limit or Requested Limit (Radiochemistry)
RPD	Relative Percent Difference, a measure of the relative difference between two points
TEF	Toxicity Equivalent Factor (Dioxin)
TEQ	Toxicity Equivalent Quotient (Dioxin)
TNTC	Too Numerous To Count

QC Association Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-24417-1

LCMS

Prep Batch: 159844

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-24417-1	Effluent	Total/NA	Water	3535	
190-24417-2	Effluent Duplic.	Total/NA	Water	3535	
190-24417-3	Field Blank	Total/NA	Water	3535	
MB 200-159844/1-A	Method Blank	Total/NA	Water	3535	
LCS 200-159844/2-A	Lab Control Sample	Total/NA	Water	3535	

Analysis Batch: 159860

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-24417-1	Effluent	Total/NA	Water	537 (modified)	159844
190-24417-2	Effluent Duplic.	Total/NA	Water	537 (modified)	159844
190-24417-3	Field Blank	Total/NA	Water	537 (modified)	159844
MB 200-159844/1-A	Method Blank	Total/NA	Water	537 (modified)	159844
LCS 200-159844/2-A	Lab Control Sample	Total/NA	Water	537 (modified)	159844

Lab Chronicle

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-24417-1

Client Sample ID: Effluent

Date Collected: 10/07/20 11:28

Date Received: 10/08/20 12:10

Lab Sample ID: 190-24417-1

Matrix: Water

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3535			159844	10/12/20 13:33	KFW	TAL BUR
Total/NA	Analysis	537 (modified)		1	159860	10/12/20 22:01	BWC	TAL BUR

Client Sample ID: Effluent Duplic.

Date Collected: 10/07/20 11:30

Date Received: 10/08/20 12:10

Lab Sample ID: 190-24417-2

Matrix: Water

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3535			159844	10/12/20 13:33	KFW	TAL BUR
Total/NA	Analysis	537 (modified)		1	159860	10/12/20 22:09	BWC	TAL BUR

Client Sample ID: Field Blank

Date Collected: 10/07/20 11:25

Date Received: 10/08/20 12:10

Lab Sample ID: 190-24417-3

Matrix: Water

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3535			159844	10/12/20 13:33	KFW	TAL BUR
Total/NA	Analysis	537 (modified)		1	159860	10/12/20 22:17	BWC	TAL BUR

Laboratory References:

TAL BUR = Eurofins TestAmerica, Burlington, 30 Community Drive, Suite 11, South Burlington, VT 05403, TEL (802)660-1990

Analyst References:

Lab: TAL BUR

Batch Type: Prep

KFW = Kurtis Wisniewski

Batch Type: Analysis

BWC = Bradley Chirgwin

Accreditation/Certification Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac Semiannual PFAS

Job ID: 190-24417-1

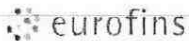
Laboratory: Eurofins TestAmerica, Burlington

All accreditations/certifications held by this laboratory are listed. Not all accreditations/certifications are applicable to this report.

Authority	Program	Identification Number	Expiration Date
ANAB	Dept. of Defense ELAP	L2336	02-25-23
Connecticut	State	PH-0751	09-30-21
DE Haz. Subst. Cleanup Act (HSCA)	State	N/A	05-16-21
Florida	NELAP	E87467	06-30-21
Minnesota	NELAP	050-999-436	12-31-20
New Hampshire	NELAP	2006	12-18-20
New Jersey	NELAP	VT972	06-30-21
New York	NELAP	10391	04-01-21
Pennsylvania	NELAP	68-00489	04-30-21
Rhode Island	State	LAO00298	12-30-20
US Fish & Wildlife	US Federal Programs	058448	07-31-21
USDA	US Federal Programs	P330-17-00272	08-09-20 *
Vermont	State	VT4000	12-31-20
Virginia	NELAP	460209	12-14-20
Wisconsin	State	399133350	08-31-21

* Accreditation/Certification renewal pending - accreditation/certification considered valid.

Eurofins TestAmerica, Michigan



Environment Testing
TestAmerica

☐ SDS or Known Hazard Information Supplied by Client

☐ Discrepancies

☐ Short Hold

☐ Rush ☐ 24 Hr ☐ 2-Day ☐ 3-Day ☐ 5-Day ☐ Other: _____

Client ID: City of Cadillac

Work Order #: 190-24417

Receipt Evaluation Performed by: Initials: Amv Date: 10/8/20 Time: 12:10

Cooler / Sample Receipt

After hours receipt: complete gray areas. Place cooler in walk-in, place form in Receiving box. Date: _____ Time: _____

Method of Shipment:

Walk-In Client Eurofins TA Field/Courier

Other Client / 3rd Party Courier: _____

Fed Ex Tracking #: _____

UPS Tracking #: 1Z E47 491 01 6045

Other: Next Day Air

Shipping Container Type:

☒ Cooler ☐ Box

☐ None ☐ Other: _____

Packing Materials:

☒ Plastic Bags ☐ Foam

☐ Bubble Wrap ☐ Paper

☐ Packing Peanuts ☐ None

☐ Other: _____

Custody Seals Intact:

☐ Yes ☐ No

☒ N/A (not used or required)

Cooling Materials:

☒ Ice (Solid) ☐ Ice (Melted)

☐ Blue Ice ☐ None

☐ Other: _____

Bacteriological Samples	Temp Corrected (°C)	Frozen?	Rec'd Within 2 Hrs?	Sample Flagged?
		Yes No	Yes No	Yes No

Received on same day sampled? Yes ☒ No

Additional Sheets Required? Yes ☒ No

Receipt Temperatures

Thermometer ID	Observed (°C)	Corrected (°C)	Temp Blank	Sample Temp	Acceptable	Cooler ID	Affected Samples
CP 313207	3.5	3.5			☒ Y ☐ N		
					☐ Y ☐ N		
					☐ Y ☐ N		

Receipt Questions**	Y	N	NA	"No" answers require additional comment
CoC present and ETA receipt signature, date, and time properly documented?	☒			
Containers and Labels in good condition? (unbroken, not leaking, appropriately filled, labels legible & attached)	☒			
Appropriate containers used and adequate volume provided?	☒			Preserved bottles checked for pH? Yes No
Number of sample containers match CoC?	☒			pH strip lot # _____
Samples received within hold?	☒			
Samples submitted for GRO and Volatiles analysis (8260, 624, 524) received without headspace?			☒	
Was a Trip Blank received with VOA samples?			☒	
Were the samples free of any questionable physical conformities? (i.e.; field duplicates or multiple bottles of the same sample do not significantly vary in appearance – color, solid proportions, etc.)	☒			
Were the CoC bottle labels and all other items free of all other discrepancies or issues that would need to be addressed with the Project Manager and/or Client?	☒			
**May not be applicable if samples are not for compliance testing				*Excludes FOG, VOAs, TOC Vials, HEM

Client Contact Record

Contact Via: ☐ Phone ☐ Email ☐ Other: _____ Person Contacted: _____ Date/Time: _____

☐ Discrepancy allowance agreement is on record in the client project file

Discussion / Resolution

28 compds per project.

Any additional documentation and clarification from the client must be noted in the narrative and/or scanned into the CoC directory.

Reviewed by: [Signature] Date: 10/8/20

WI-MI-010_020720

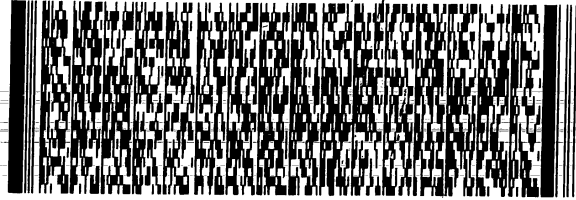
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TO **SAMPLE RECIEVING**
EUROFINS TEST AMERICA BURLINGTON
30 COMMUNITY DR
STE 11
SOUTH BURLINGTON VT 054036834

(802) 660-1990
INV:
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